Data Profiling Report

Introduction:

Data profiling is the process of examining a dataset to understand its structure, quality, completeness, and consistency before performing analysis. The purpose of this report is to identify data quality issues and document the cleaning process applied to prepare the dataset for SQL analysis and Power BI dashboard development.

Dataset Overview:

Dataset Name: ERP Sales Dataset

File Format: CSV

Number of Records: 9

Number of Columns: 27

Domain: Sales and Customer Transactions

Data Quality Assessment:

The dataset was analyzed to identify data quality issues before further processing. The following issues were found:

- Duplicate records were present.
- Null (missing) values were found in multiple columns.
- Some customer email addresses were invalid.
- Country names were inconsistent (e.g., Pakstan, UAE, KSA).
- Negative values were found in the Quantity column.
- Zero or negative values existed in the UnitPrice column.
- Extra spaces and inconsistent text formatting were identified in several text fields.

Data Cleaning Process:

The following data cleaning operations were performed using Microsoft Excel Power Query

- Removed duplicate records.
- Trimmed leading and trailing spaces.
- Applied Clean transformation to remove unwanted characters.
- Converted Customer Name values into Proper Case.
- Standardized country names.
- Removed invalid email records.
- Filtered negative Quantity values**.**
- Removed zero and negative UnitPrice values.

- Added calculated columns:
- TotalSales
- OrderYear
- OrderMonth

Data Quality Improvement:

After performing data cleaning:

- Duplicate records were eliminated.
- Invalid records were removed.
- Missing and inconsistent values were handled where applicable.
- Text formatting became standardized.
- Numeric values were validated.
- The dataset became consistent and ready for database creation and business intelligence reporting.

Conclusion:

The data profiling and cleaning process significantly improved the overall quality of the dataset. The cleaned dataset is accurate, consistent, and suitable for SQL database implementation, analytical queries, and Power BI dashboard development.